

Week 3 Simulator

Introduction to Wireless Communications

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Code explanation:

The Week 3 code extends the Week 2 path loss simulator by importing its results directly and adding log-normal shadowing and outage probability analysis. A random Gaussian variable with $\sigma = 8$ dB is generated and added to the received power at each distance point, simulating unpredictable real-world signal variations caused by obstacles like buildings and trees in urban environments.

Two new plots are produced — received power with and without shadowing, and outage probability vs distance using the Q-function formula. Results confirm that higher frequencies like 28 GHz become unreliable much sooner, crossing the 10% outage threshold at a fraction of the distance compared to 2.4 GHz, which directly explains why mmWave 5G networks require much denser cell tower deployment in practice.